Unit 20 – Client Side Customisation of Web Pages
Assignment 2 – Website Scripting

Introduction

As part of your continued internship at InsertName Website Development you have been asked to produce a document that outlines javascript, its uses and how it can improve page functionality. You have also been asked to include some information about some of the problems web developers may face when creating sites that will be viewed on a number of different browsers.

(P3) Explain the fundamentals of Javascript

Javascript Introduction

Javascript is a scripting language which is client-sided and the programming language is mostly used by web browsers and it is performed by a browser to add extra functionality to a web page. Javascript can be used to manipulate HTML and CSS and it is commonly used to validate web forms and perform calculations in the browser.

Javascript differs from HTML and CSS because they aren't programming languages, while Javascript is a programming language. Javascript also builds the behaviour of the web site, while HTML is the structure and CSS changes the style.

Characteristics of Javascript

- Event Driven

Event driven programming is important for Javascript because it gives extra user functionality on the website, hence why Javascript is centered on event driven programming. Event driven programming is when component takes events and triggers them.

For example, actions such as mouse clicks, key presses, sensor outputs or messages from other programs are all events in event driven programming. Basically, in an event-driven application, there is a main loop that listens to events, and then triggers a callback function when one of those events are detected.

- Objects

Objects is probably the most important part of Javascript to understand because if you understand objects, you would understand Javascript. This is because most things in Javascript are objects, some tutorials or people may say "Everything in Javascript is an object", but it's not exactly 100% true.

The reason it's not 100% true but mostly true is because instead of objects there are primitives, but sometime primitives can be objects too. Objects in Javascript consist of: Dates, maths, regular expressions, arrays, functions and, of course, objects you create. Other values in Javascript can also be objects or are primitive data that are treated like objects, such as booleans, numbers and strings.

- Methods

In Javascript, a function is a named block of code that can be claimed from anywhere within the code. The difference between a function and a method is basically what programming language is involved so they practically mean the same thing. Methods eliminates the need of writing the same code again and again and it allows a programmer to divide a big program into a number of small and manageable functions.

- Handling Events

Handling events is the programs which work with direct user input, such as a keyboard or mouse input. User interaction is unpredictable therefore this Javascript characteristic requires a different approach to control flow. For example, event handlers, which is an event that tracks user interaction such as click, double clicks, hovers, keyboard hits, etc.

- Security Issues (e.g. opening/closing windows)

Javascript does a few security risks and issues to the user because it’s client-sided. For example, a risk that can affect the client is that their cookie can be stolen so that another user can impersonate them. Another common method of attack is to inject Javascript into the page via a form submission.

- Inclusion in HTML

Javascript can be included in HTML but HTML cannot reference a specific JS function. In the HTML code you include a script with a “<script>” element that accesses whatever parts of the DOM (Document Object Model, which is a platform and language-neutral interface that allows programs and scripts to access and update the content, structure and style of the document) you want it to interact with via the “document” object that the browser will make available on the script.

<u>Uses of Javascript</u>

- Alerts

The alert box is a small dialog box that is displayed on the screen of the user to give information or to warn the user about an operation which could potentially damage the system. For example, it can be used for warning the user if the system is deleting one or more files or an action cannot be done.

- Confirming Choices

Confirming choices is a lot like the alert box, because it is a small dialog box that is displayed on the screen of the user. The difference of the confirming choices box is that the box is used for verifying or accepting something, instead of a warning for an error. When a confirming choices box pops up, the user will have to either click OK or Cancel to proceed.

- Browser Detection

Browser detection is a code which its purpose is of checking whether or not the user has a certain browser capability such as new features on HTML5 or the type of browser and version that is being used, for example Firefox 50.0 Windows 10.

- Validation

Validation is a number of text boxes which is required to be filled in in order to get information from the user. It is used mostly in registration forms to gather information of the user so they can access a website that has some sort of restriction or to verify/create an account for a website.

Constructs of Javascript

- Values

Values in Javascript is neatly separated data which plays its own role. What I mean by this is that every value has a type, which determines the kind of role it can play. For example, there are six basic types of values: Numbers, strings, booleans, objects, functions and undefined values.

- Variables

Variables in Javascript acts like a container for data. What I mean by this is that variables contains data which can be referenced anywhere within the script simply by naming the container. Below is an example of a Variable code:

```
var price1 = 5;
var price2 = 6;
var total = price1 + price2;
```

- Assignment

Assignments in Javascript are operators that assign a value to the left operand based on the value of its right operand. What I mean by this is that the shorthand operator of “x = y” would equal x = y. A better example would be the subtraction assignment, which looks like “x -= y”, means x = x - y. This also applies to addition, multiplication and division.

```
var x = 10;
x += 5;
```

- For loop

For loops in Javascript is basically the command for creating a loop. What I mean by this is that a sequence of instructions is going to be continually repeated until a certain process is done, such as getting an item of data and changing it. Below is an example of a For Loop code:

```
for (i = 0; i < cars.length; i++) {
    text += cars[i] + "<br>";
}
```

- While Loop

While loops in Javascript is a loop that executes a specified statement as long as the loop evaluates to true. What I mean by this is that is the specified statement will be executed when the evaluation is true, if it is false, the loop will stop. Below is an example of a While Loop code:

```
while (i < 10) {
    text += "The number is " + i;
    i++;
}
```

- If statement

If statement in Javascript is used to specify blocks of code to be executed, but only if the condition is true. What I mean by this is that the If statement will only work if it's true to the block of code, if it's false it won't be executed because that would be the Else statement to execute a false condition. Below is an example of an If statement code:

```
var x = document.getElementsByTagName("DIV")[0];

if (x.id === "myDIV") {
    x.style.fontSize = "30px";
}
```

- Functions

Functions in Javascript is used to perform a particular task and it's only executed when something refers to the block of code. What I mean by this is that it can be called by code external or internal to the function. Functions are first-class objects, this means that they can have properties and methods just like any other object. Functions are like procedures but they are not the same, a functions will always return a value, and a procedure might not. Below is an example of a Function's code:

```
function myFunction(p1, p2) {
    return p1 * p2;              // The function returns the product of
p1 and p2
}
```

- onFocus

OnFocus in Javascript is when an element gets focus when an event occurs. What I mean by focus is that anything that is triggered with focus, it can change the element, i.e. colour. However, not all elements are focusable by default. Below is an example of an onFocus code:

```
object.onfocus=function(){myScript};
```

- onLoad

OnLoad in Javascript is when an object has been loaded when an event occurs. What I mean by this is that the code will be executed as soon as the page has been loaded. This code can be used to check the browser type and version of the user, which would load the proper webpage based on the information gathered about the browser. Below is an example of an onLoad code:

```
object.onload=function(){myScript};
```

- click()

Click() in Javascript is when a mouse click is simulated on the method. What I mean by this is that the method can be used to execute a simulation of a click on an element as if the user manually clocked the element itself. Below is an example of a code with click():

```
<input type="checkbox" id="myCheck" onmouseover="myFunction()"
onclick="alert('clicked')">

<script>
// On mouse-over, execute myFunction
function myFunction() {
    document.getElementById("myCheck").click(); // Click on the
checkbox
}
</script>
```

- Properties (of an object)

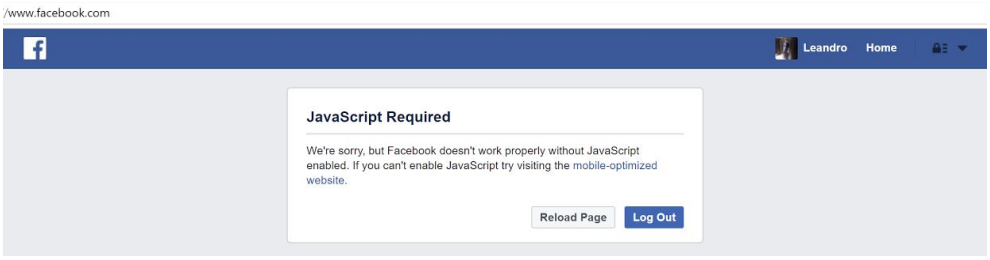
Object properties in Javascript is a collection of unordered properties. What I mean by this is that there are different properties of objects that can usually be added, deleted and changed. Properties are very important in Javascript object because they are the values associated with a Javascript object. Below is an example of properties of an object:

```
person.firstname + " is " + person.age + " years old.";
```

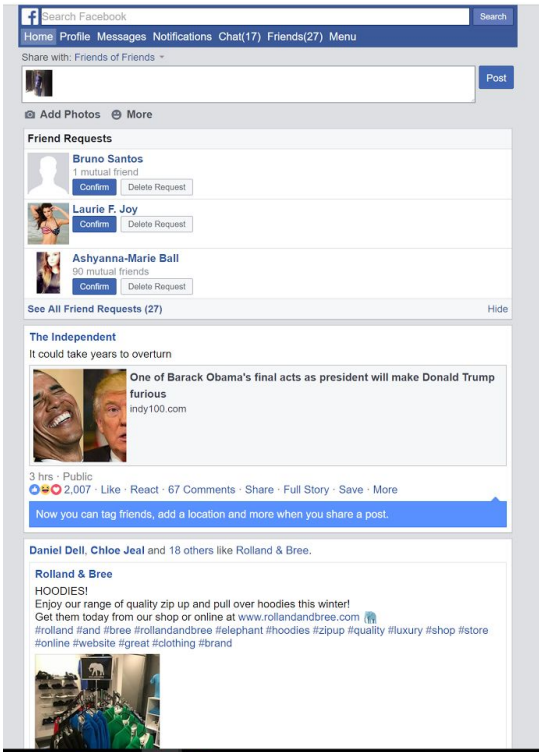
(M2) Discuss how a scripting language can improve functionality.

In this part of the assignment I will be checking how scripting languages have enabled the features that I can see as part of Facebook.

When I turned JS off, I noticed that I had a warning message on my screen saying “Javascript Required” and that facebook wouldn't work correctly without JS and I was forced to launch facebook in the “mobile-optimized website”.



Below is a screenshot of the mobile-optimized website of Facebook.

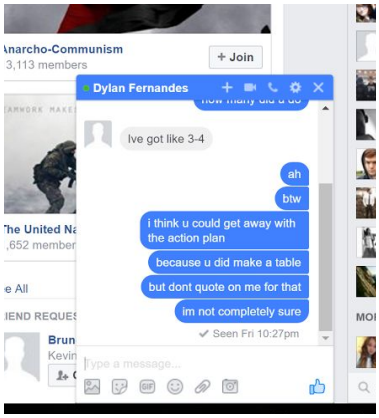


Below are the changes I have noticed with JS on its default setting (on) and with JS off.

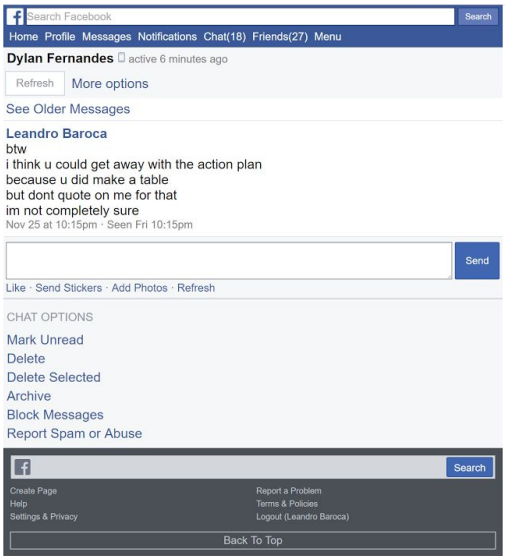
- Chat

As I could see when I use the default browser settings (with JS on), I noticed that facebook and the chat box were working ‘normally’ or how it should be. By that I mean that the chat box was opening at the bottom right, the chat was refreshed automatically

without me needing to manually refresh, it says “seen” with a tick, the time and date it was seen when the user opens the chat box with the message I sent in it and a small speech bubble appears when the user on the other side is sending a message.



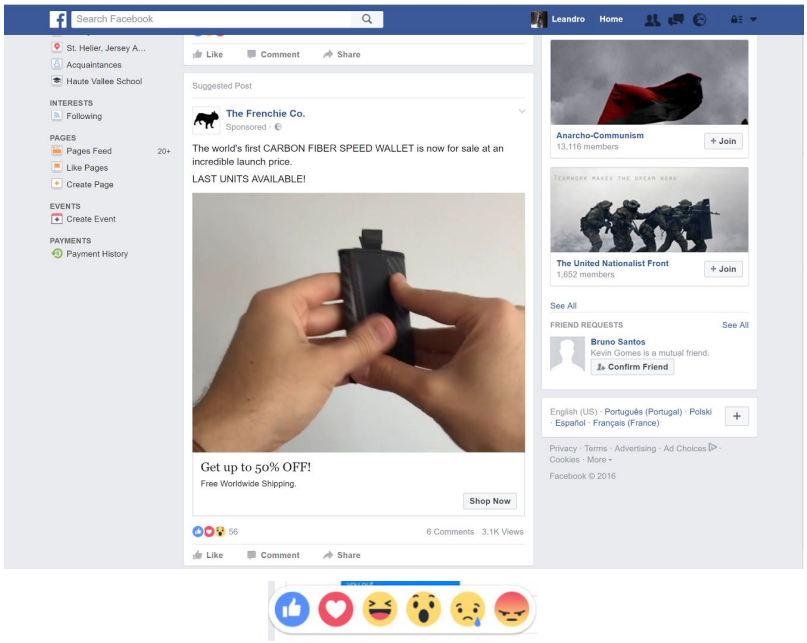
After opening the “mobile-optimized website”, I noticed that the interface was very different to that on the default website with JS support. I have noticed that the profile pictures have disappeared, the chat is no longer live (meaning I have to manually refresh the page to see new messages) and that the ability to multitask is taken way (meaning that I would have to go to a different page to look at the news feed, etc).



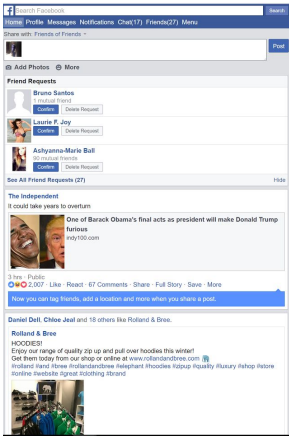
In conclusion, The most significant improvements that Javascript has done to the chat is that Javascript does automatic refreshing to the live chat (instead of manually refreshing it), with JS it looks much more simple and modern, less cluttered, faster accessibility and the ability to multitask.

- Newsfeed

As I could see with the default browser settings on (with JS on), I could see that the newsfeed worked correctly with the normal, modern interface and the refresh of new newsfeeds appearing. I also noticed the reactions popping up from hovering the like buttons and the comments box opening when I clicked on it.



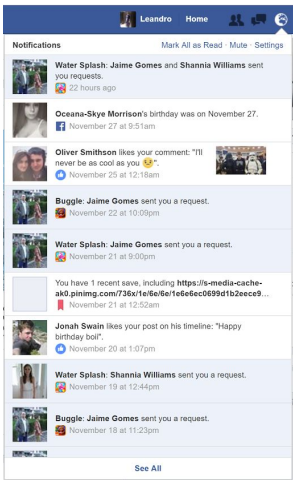
When I turned off JS and went on the “mobile-optimized website”, I noticed that the modern and simplistic look of facebook was gone and it was all very basic, when I scrolled down I noticed that it did not automatically refresh the older newsfeed and I had to manually refresh it by clicking the “see more” button. I also noticed that the reactions did not pop up and the comment box opened up in a different way to that on the website that supports JS.



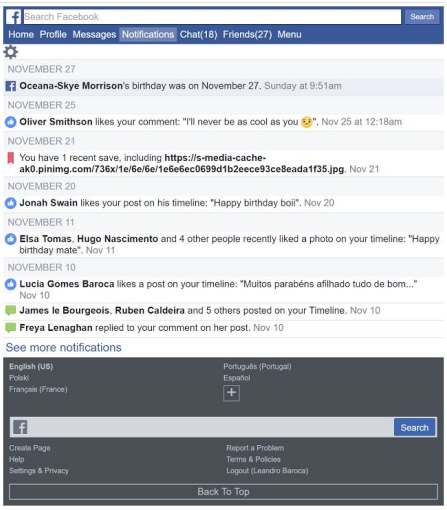
In conclusion, Javascript makes significant functionality improvements such as an automatic refresh of new and old newsfeed, the modern and simplistic look, profile pictures, reactions created from hovering over the like button, the common box appearing differently and all the links are like hyperlinks (blue and underlined when hovered over it).

- Notifications

With the default browser on (JS on), I could see that the notification bar is filled with Javascript. For example, the notification icon changes into a white icon (dark blue being the default), the box appears when the icon is clicked, the profile pictures of users show up, the notifications that have not been viewed and the notification icon has a little number on it showing the amount of unseen notifications (unseen in the screenshot).



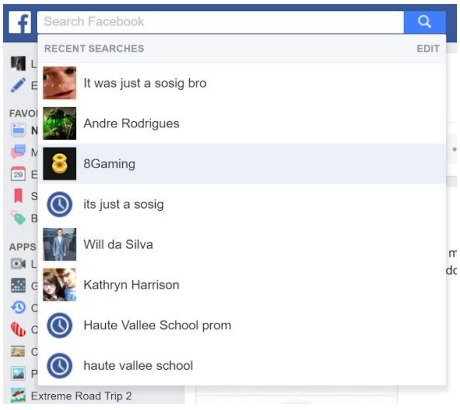
When I turned JS off I first noticed that the notification was no longer a box but a separate page and that the profile pictures were not appearing. There was also no longer an icon for the notifications and there isn't an ability to scroll down because there isn't any auto refresh that Javascript has the ability to do, there isn't also an indication for unread notification like the default one.



In conclusion, I noticed that Javascript made the notification box a lot more functionable because it's easier to access the notification box and the interface is much more modern and simplistic (which makes it easy on the eyes and less cluttered), the unread notifications appear as a darker square (making it easier to see what you have and haven't seen) and that it shows the profile pictures of people so it's easier to spot out who the notification is from and quicker if you are a slow reader.

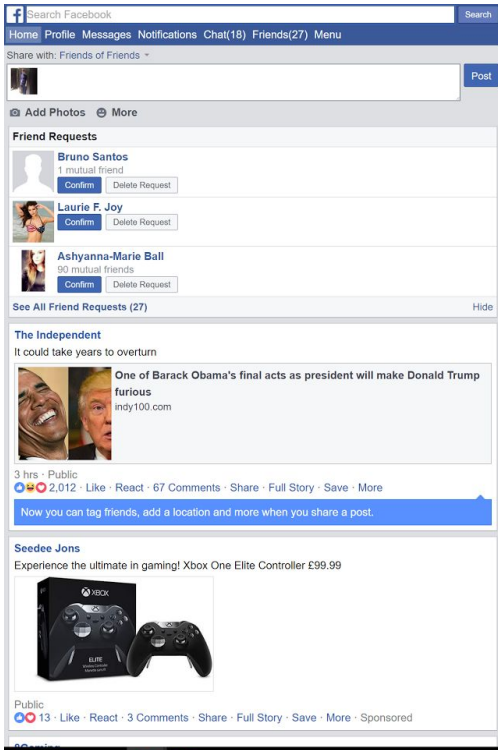
- Search bar

With Javascript turned on in the browser, I noticed that the search bar worked like it was intended to. By this I mean that when I clicked on the search bar a box appeared on the bottom with my last searches and I can click on any of them to visit the page. I have also noticed when I hover the mouse over the pages, the box changes colour as the example shows below.



When I turned Javascript off I didn’t notice many visual changes with the search bar, it still looked very similar to the Javascript one, it still has the “Search Facebook” text, it is next to the facebook logo, it's on the top of the newsfeed page and overall very minor

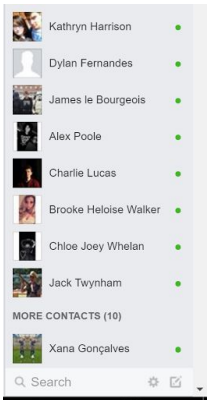
differences. But when I interacted with it I noticed that a box didn't appear with my last searches and when I typed something in the search bear the drop down box didn't appear with the related or recommended searches.



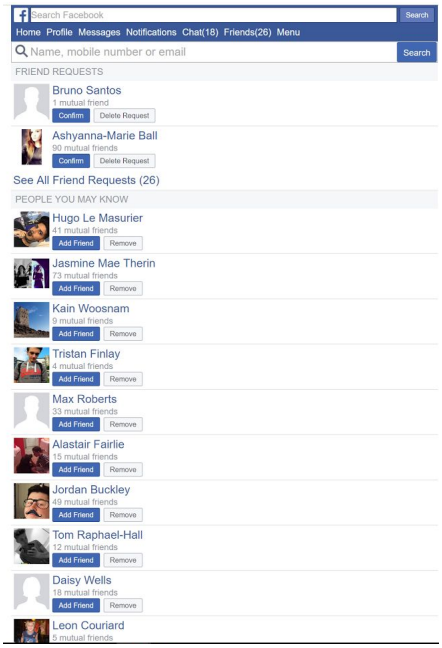
In conclusion, I noticed that Javascript made it a lot easier and less time consuming to search for something you want or to look for something which you have searched before. Without Javascript you wouldn't be able to look at previous or recommended searches and it would be very time consuming.

- Friend search

On the default browser with Javascript enabled I noticed that the friend search was a small box on the bottom right of the screen, which was easily accessible and when I searched a name it would narrow down the the person that included the characters of what I typed on, clicking on their name also opens a box which is the chat box mentioned earlier. I also noticed that it shows the friends that are online and the ones that aren't online.



When I turned Javascript off, I noticed that the friend search was in a different page instead of the bottom right of the news feed screen. On the friends page I noticed that there was a search bar and when I typed something in, the amount of friends didn't reduce to the ones who had matching characters and a box didn't appear, instead I had to click enter or search and it refreshes the page showing the people with the matching characters. I also noticed that it doesn't show if the friends are online or not.



In conclusion, the Javascript made it a lot easier and less time consuming to find friends because of the convenient placement of the bottom right of the screen and the ability to search for friends in significantly faster speeds than that compared to the browser with JS disabled. I was also surprised to see that the profile pictures showed up on the friends tab because I have never seen profile pictures in the browser with JS disabled.

(D1) Explore how web pages using scripts are implemented in different browsers.

Why is it necessary to be able to run different scripts depending on a user's browser type?

It's necessary to be able to run different scripts depending on the type of browser because all the browsers work differently because of their unique engines. For example, there's Google Chrome and Opera with the 'Blink' engine, Internet Explorer with the 'Trident' engine, Firefox with the 'Gecko' engine, Safari with the 'WebKit' engine, Opera with the 'Presto' engine and the new Microsoft Edge with the 'EdgeHTML' engine.

The reason that Javascript is different depending on the user's browser is because there isn't a unified Javascript engine. So the browsers can do whatever they wanted in terms of making up the rules for which to access and manipulate JS, CSS and HTML elements in a web page.

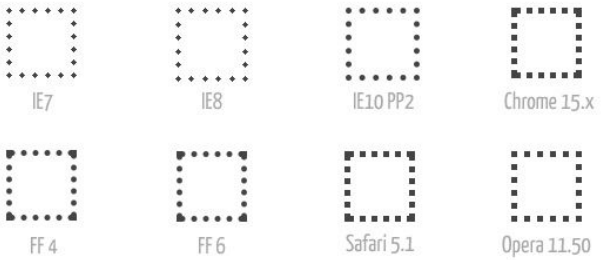
That is the reason it is necessary to be able to run different scripts depending on a user's browser type, because if you use a script sample from Google Chrome and use it for Mozilla Firefox it would be very likely that the script code would not work and cause an error in the Firefox browser. This isn't only applied to Firefox too, it can be applied in other browsers that aren't Google Chrome.

How does CSS differ between browsers? (using border styles as an example.)

Not only does Javascript have a difference between browsers, CSS has a noticeable difference between browsers too. For example, the simple border property in CSS, it varies on every browser that exists to this day.

The differences include the "dashed" and the "dotted". There are other border styles like the double, groove, ridge, inset and the outset which are also different but they are too similar or too small of a difference for comparison.

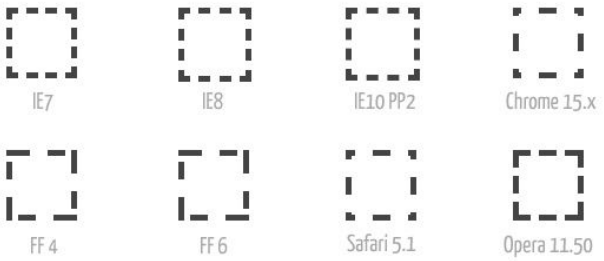
For this example I will be using the dotted and dashed borders because of the significant difference the browsers have. The first one is the dotted:



As you can tell, some of the dotted look like circles and some look like diamonds, but Firefox, Chrome and Safari have very strong corners. IE10 is the only browser that has

full circular dots that go all the way around. Even though Opera has a neat looking set of squares, so does Safari and Chrome with their squares but with corners, it is not the correct way of making a dotted border because it should be “a series of round dots.”

Secondly, I will be showcasing the differences in the dashed border:



As you can tell, Chrome, Safari and Firefox has larger spacing between the gaps, causing them to render differently from the others. The most noticeable one is Firefox with its unusual way of not having the dashes wrapping in the corners. Again, opera does look the neatest and the IE versions are very basic dashed borders.

How can you implement different stylesheets on the different browsers?

When using CSS you would probably notice that it could only be used on the browser you’re making it for, i.e. Google Chrome, and it wouldn't work with others (However, it will work for most versions of IE such as IE6, IE7 and IE8). The reason for this is that, unfortunately, a cross browser solution does not exist if you are just doing it on the HTML.

So a solution for this is to use the Javascript “Browser Detection”. It does not only just detect the type of browser, it also knows the version of the browser, what OS the browser is on. For example, if you are trying to use Mozilla Firefox, you would just type in: ‘if (BrowserDetect.browser.indexOf(“mozilla”)>-1) { document.write(‘<’+’link rel=”stylesheet” href=”../component/mozilla Styles.css” />’);’.

However, if you want to include more browsers to implement the stylesheets to, you just do the same code but instead of ‘if’, you replace it with ‘else if’, which can be repeated for as many times as you like. Obviously you also include the different browsers such as ‘explorer’ and put in ‘explorerStyles.css’.

How does javascript differ between Firefox and IE? (pick 3 examples)

In Firefox and Internet Explorer, Javascript has many different varieties, this is due to the fact different browsers have different engines. So it’s no surprise that Javascript varies from Firefox and IE, I will be listing 3 examples of how Javascript varies from both these browsers.

The first example is the accessing the “class” Attribute of an Element. The two browsers use different syntax to access this attribute via the ‘getAttribute’ method in Javascript.

For example, the IE syntax would have code similar to ‘var myObject = document.getElementById(“header”); var myAttribute = myObject.getAttribute(“className”);’.

While the Firefox syntax would have similar code to ‘var myObject = document.getElementById(“header”); var myAttribute = myObject.getAttribute(“class”);’. This syntax could also apply when using the ‘setAttribute’ method.

The second example is the accessing the “for” Attribute of the <label> Tag. Similar to the first example, there are different syntax to access a <label> tag’s “for” attribute in Javascript.

For example, the IE syntax would have code similar to ‘var myObject = document.getElementById(“myLabel”); var myAttribute = myObject.getAttribute(“htmlFor”);’.

While the Firefox syntax would have similar code to ‘var myObject = document.getElementById(“mylabel”); var myAttribute = myObject.getAttribute(“for”);’.

Lastly, the third example is the CSS “float” property. The basic syntax for accessing a specific CSS property for a given object is ‘object.style.property’.

For example, the IE syntax would have code similar to ‘ document.getElementById(“header”).style.styleFloat = “left”;’.

While the Firefox syntax would have similar code to ‘document.getElementById(“header”).style.cssFloat = “left”.

How can you execute different javascript code on different browsers?

As mentioned before, different browsers have different engines which means that some of the Javascript is going to vary between the browsers. Of course, executing the code on different browsers are going to be different is some aspect as well.

For example, like I have mentioned before between the different comparisons of IE and Firefox, those 2 browsers aren't the only ones who differ from each other. If you copy one of IEs or Firefox’s code to another browser such as Chrome or Opera, it would most likely not execute and give you an error.

There is a way to overcome the barrier of this because the different browsers have different engines, of course, but Javascript doesn't have a specified engine of it’s own. So it isn't impossible to overcome the barrier of the different engine.

One way of using different JS code on different browsers is the User agent string. Even though this method is thought to be unreliable, because it’s too insignificant to trick this value.

Another way of overcoming this barrier is to use the browser detection method, but only when it’s truly necessary. Such as showing browser-specific instructions to install an extension.
The recommended method is to use the feature detection when possible. The reason for this is because feature detection has proven to be a more effective technique that requires less maintenance.